

Solve Any Data Analysis Problem

Chapter 5 - Project 4 - Example solution - Part 2

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

import datetime
```

Definitions

We need to define "pre-", "during" and "post-" COVID years in our data. "Post" in this case will mean when lockdown restrictions were lifted, not when the pandemic ended, which is much more debatable:

```
In [2]: COVID_START_DATE = datetime.datetime(2020, 3, 1)
LOCKDOWN_END_DATE = datetime.datetime(2021, 7, 1)
```

Admissions

```
In [3]: admissions = pd.read_csv("admissions.csv")
print(admissions.shape)
admissions.head()
```

(48, 3)

```
Out[3]:
```

	Year	Month	Admissions (million)
--	------	-------	----------------------

0	2018	January	16.2
1	2018	February	16.1
2	2018	March	13.5
3	2018	April	15.5
4	2018	May	13.7

```
In [4]: admissions["date"] = (
    "1 " +
    admissions["Month"] +
    " " +
    admissions["Year"].astype(str)
)

admissions["date"] = pd.to_datetime(admissions["date"], format="%d %B %Y")
admissions.head()
```

```
Out[4]:
```

	Year	Month	Admissions (million)	date
0	2018	January	16.2	2018-01-01
1	2018	February	16.1	2018-02-01
2	2018	March	13.5	2018-03-01
3	2018	April	15.5	2018-04-01
4	2018	May	13.7	2018-05-01

Mark pre-, during and post-lockdown dates

```
In [5]: admissions.loc[admissions["date"] < COVID_START_DATE, "covid_period"] = "pre-COVID"
admissions.loc[admissions["date"].between(COVID_START_DATE, LOCKDOWN_END_DATE, "left_inclusive"), "covid_period"] = "during COVID"
admissions.loc[admissions["date"] >= LOCKDOWN_END_DATE, "covid_period"] = "post-lockdowns"

# also make it a Categorical data type to be able to specify the order
admissions["covid_period"] = (
    pd.Categorical(
        admissions["covid_period"],
        categories=["pre-COVID", "during COVID", "post-lockdowns"],
        ordered=True
    )
)

admissions["covid_period"].value_counts(dropna=False)
```

```
Out[5]: pre-COVID      26
during COVID    16
post-lockdowns   6
Name: covid_period, dtype: int64
```

```
In [6]: fig, axis = plt.subplots(figsize=(10, 6))

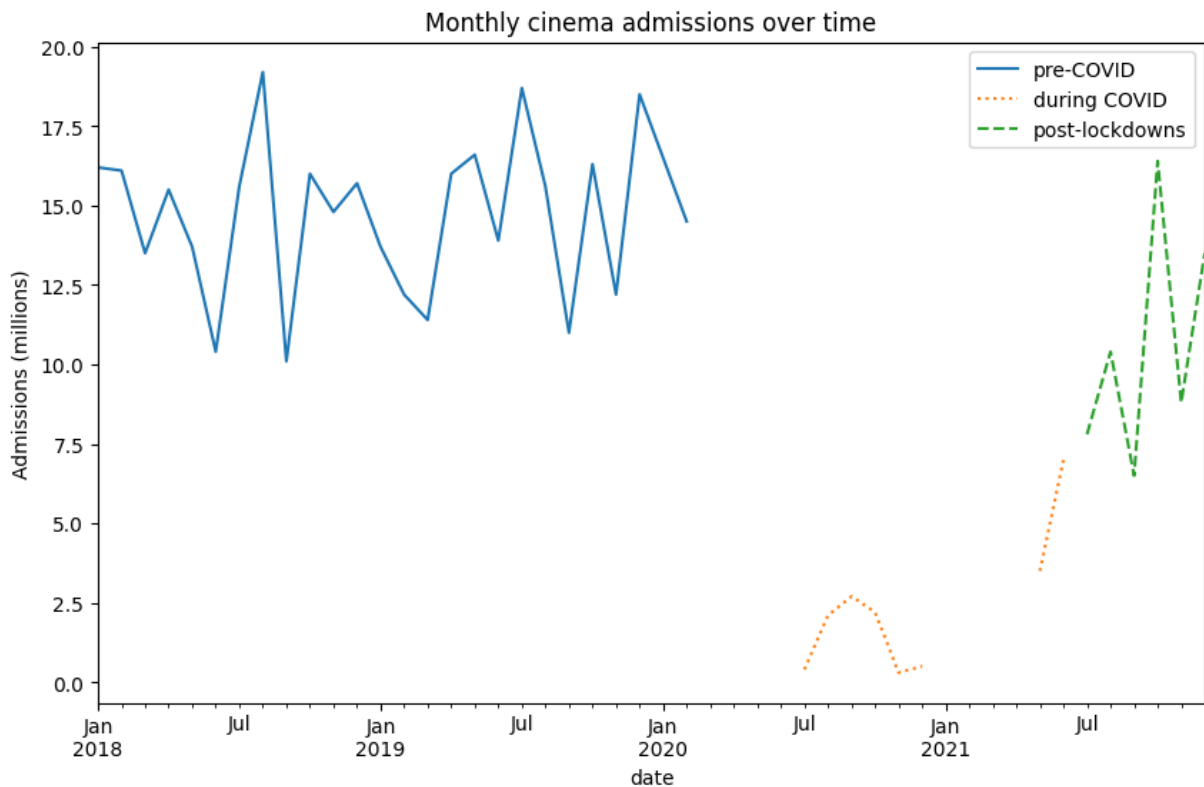
linestyles = ["solid", "dotted", "dashed"]

for idx, covid_period in enumerate(admissions["covid_period"].value_counts().index):
    (
        admissions
        .query(f"covid_period=='{covid_period}'")
        .set_index("date")
        ["Admissions (million)"]
        .plot(ax=axis, label=covid_period, linestyle=linestyles[idx])
    )

axis.set(
    title="Monthly cinema admissions over time",
    ylabel="Admissions (millions)"
)

axis.legend()

plt.show()
```



You can definitely see the covid impact!

```
In [7]: fig, axis = plt.subplots()

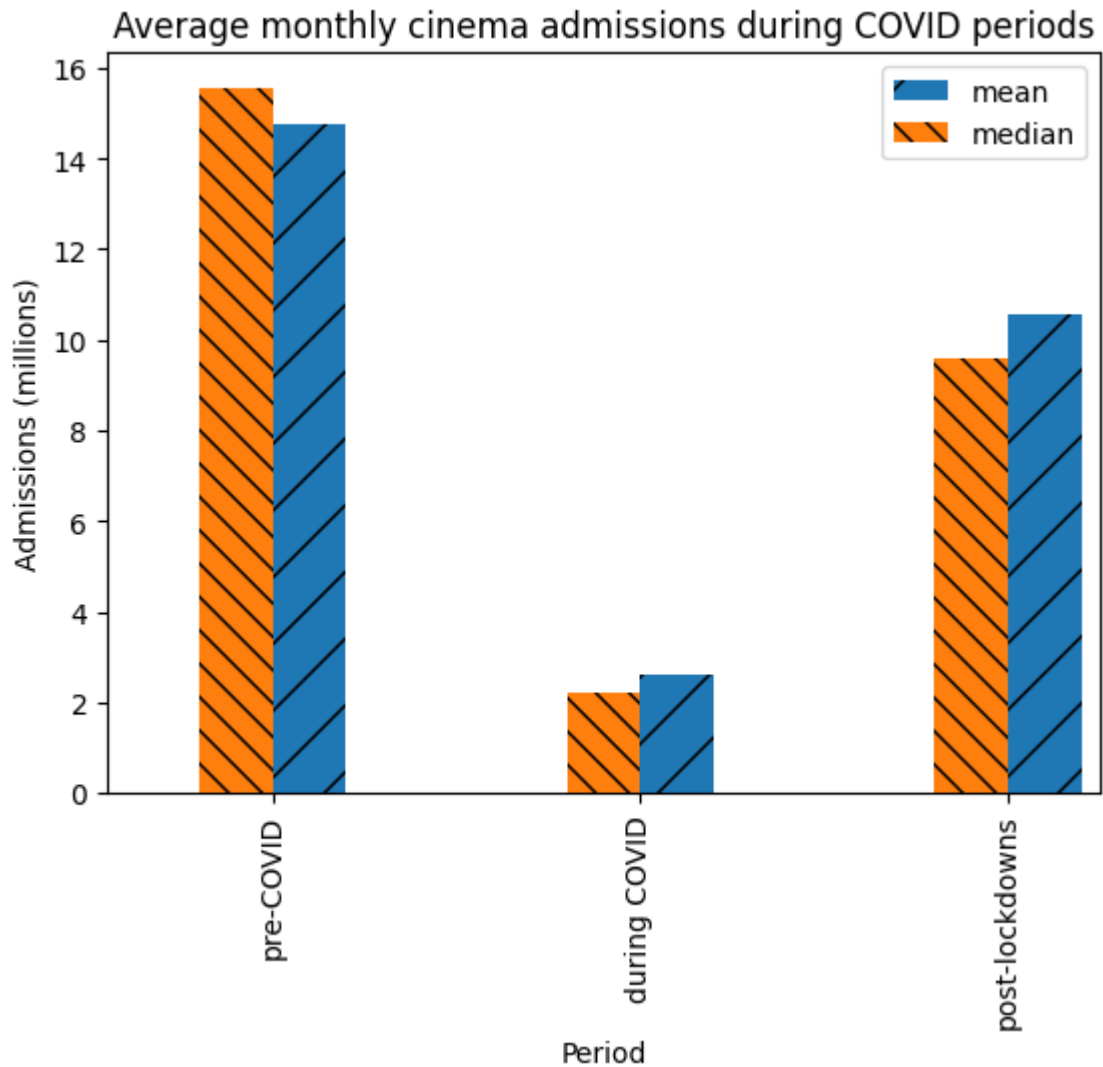
admissions_by_period = (
    admissions
    .groupby("covid_period")
    ["Admissions (million)"]
    .agg(["mean", "median"])
)

# Create a bar plot for each metric with different hatch patterns
for i, metric in enumerate(admissions_by_period.columns):
    hatch = "/" if i == 0 else "\\\\" # Different hatch patterns for different met
    color = "C0" if i == 0 else "C1" # Different colors for different metrics
    admissions_by_period[metric].plot(kind="bar", ax=axis, position=i,
                                       hatch=hatch, label=metric,
                                       width=0.2, color=color)

axis.set(
    title="Average monthly cinema admissions during COVID periods",
    ylabel="Admissions (millions)",
    xlabel="Period",
    xticklabels = admissions_by_period.index
)

axis.legend()

plt.show()
```



The average monthly attendance is definitely lower, another indication that things haven't returned to pre-COVID levels.

Interestingly the mean and median are reversed post-lockdowns, let's dig in a little bit more:

```
In [8]: fig, axes = plt.subplots(1, 2, figsize=(10, 6), sharey=True)

(
    admissions
    .loc[admissions["covid_period"] == "pre-COVID", "Admissions (million)"]
    .hist(bins=10, ax=axes[0])
)

axes[0].set(
    title="Distribution of monthly admissions pre-COVID",
    xlabel="Admissions (million)",
    ylabel="Frequency"
)

(
    admissions
    .loc[admissions["covid_period"] == "post-lockdowns", "Admissions (million)"]
```

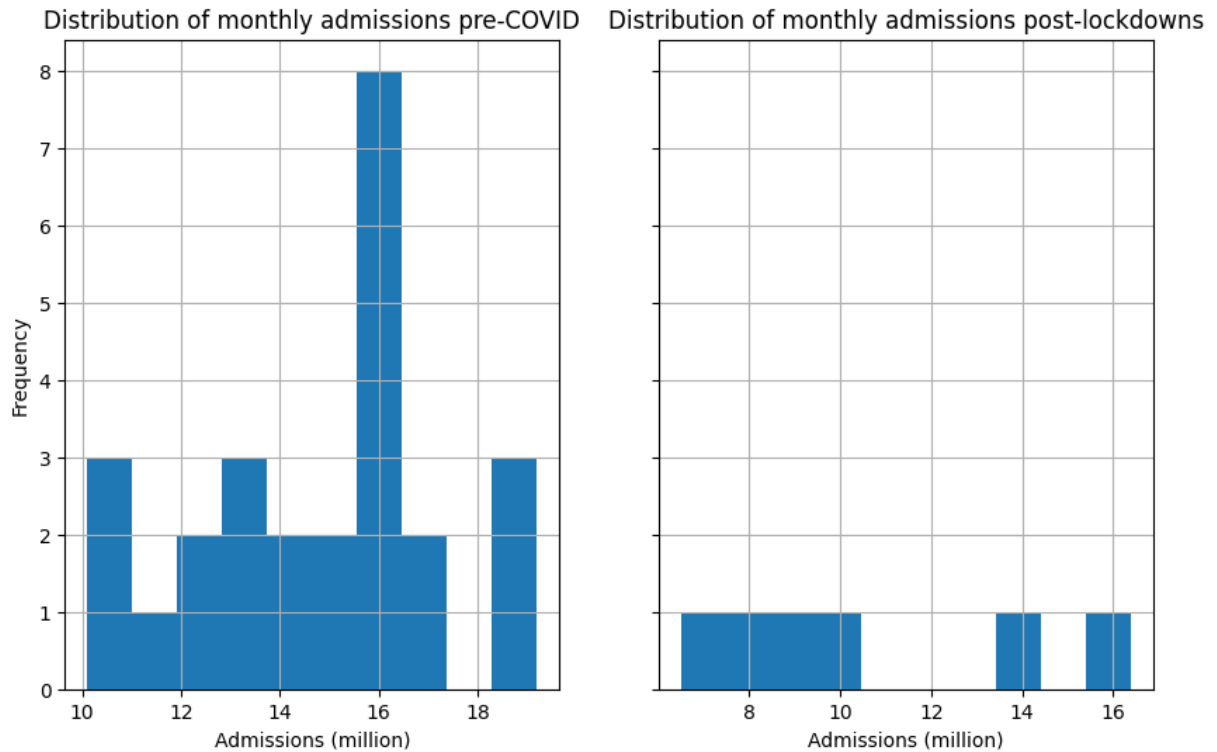
```

    .hist(ax=axes[1])
)

axes[1].set(
    title="Distribution of monthly admissions post-lockdowns",
    xlabel="Admissions (million)"
)

plt.show()

```



We have less data post-lockdowns and because of the low initial numbers we have more of a right skew (pre-COVID we had more of a left skew, where lower numbers were more rare).

Finally, let's look at which months are most popular pre-COVID and post-lockdowns. Perhaps people's admission patterns have changed.

```

In [9]: fig, axes = plt.subplots(3, 1, figsize=(6, 12))

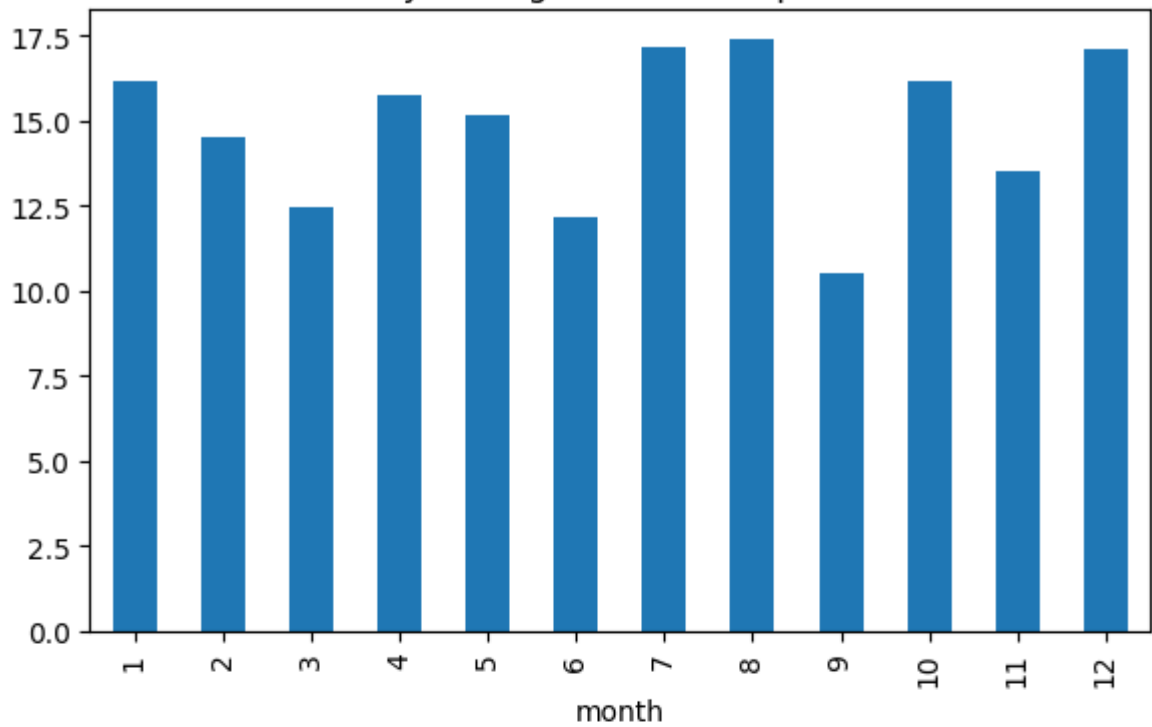
for idx, period in enumerate(admissions["covid_period"].unique()):
    (
        admissions.assign(month=admissions["date"].dt.month)
        .query(f"covid_period == '{period}'")
        .groupby("month")
        ["Admissions (million)"]
        .median()
        .plot
        .bar(ax=axes[idx])
    )

    axes[idx].set(
        title=f"Monthly average admissions {period}"
    )

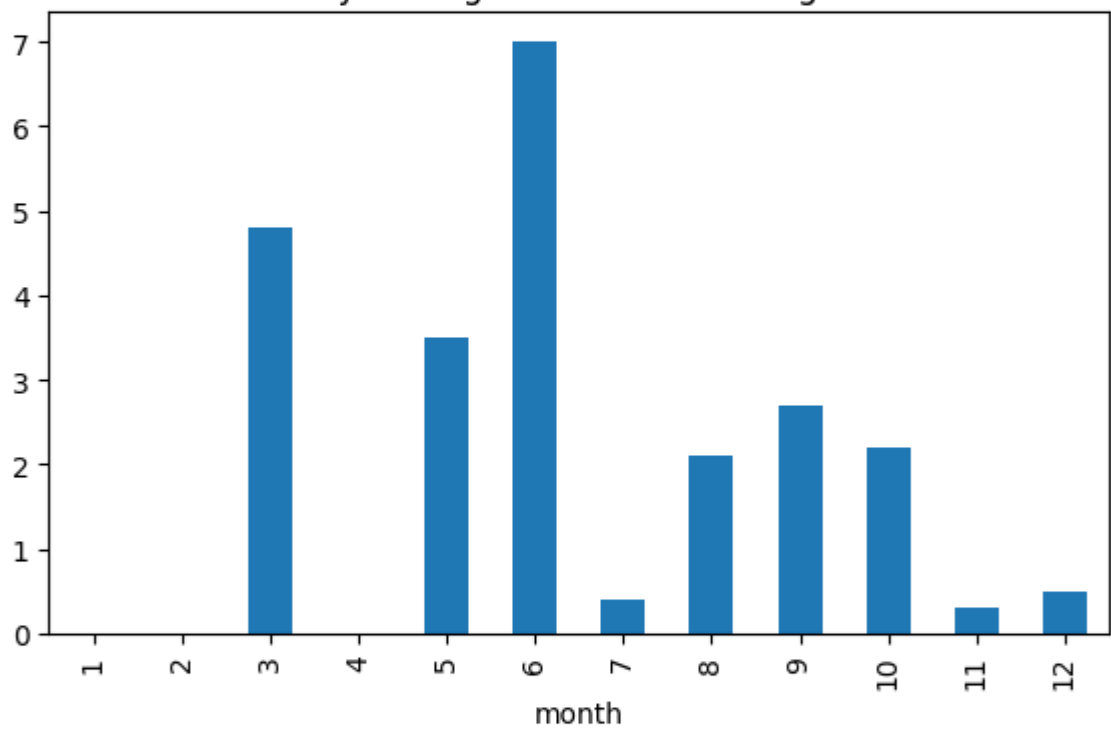
```

```
plt.tight_layout()  
plt.show()
```

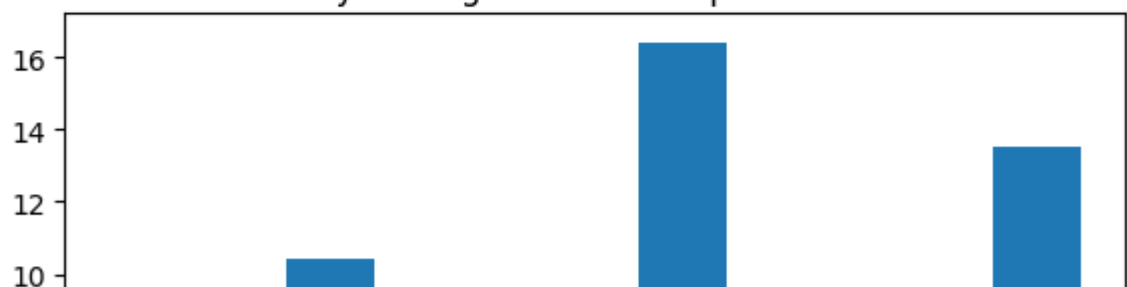
Monthly average admissions pre-COVID

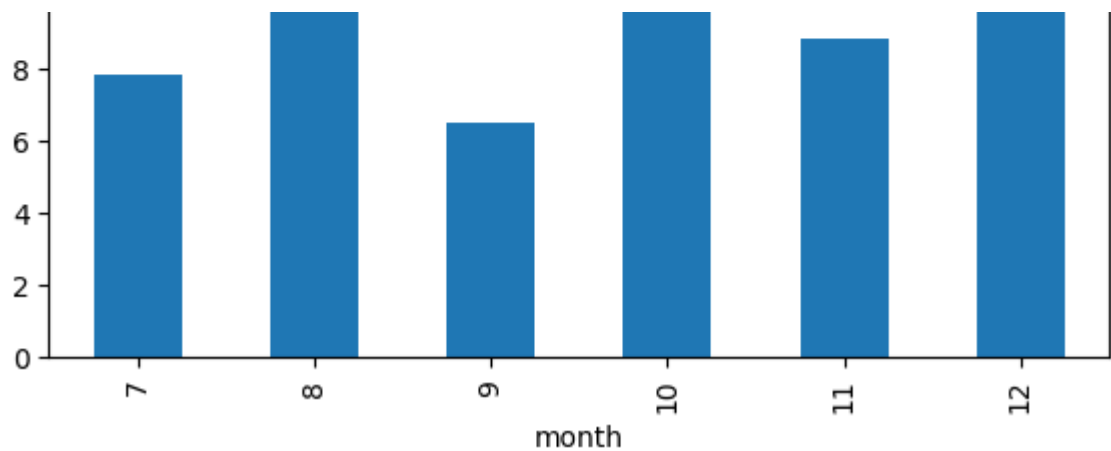


Monthly average admissions during COVID



Monthly average admissions post-lockdowns





Genres

```
In [10]: genres = pd.read_csv("genres.csv")
print(genres.shape)
genres.head()
```

(67, 5)

```
Out[10]:
```

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
0	2018	Action	77	361.3	Avengers: Infinity War
1	2018	Animation	46	242.2	Incredibles 2
2	2018	Drama	246	154.1	A Star Is Born
3	2018	Comedy	144	88.3	Johnny English Strikes Again
4	2018	Biopic	5	78.5	Bohemian Rhapsody

Need to define COVID years a bit differently because our data is annual, not monthly.

Since we're interested in pre- vs. "post", let's just categorize using those two labels. We need to make a call on 2021. Does it count as post-COVID? We're interested in the second half of the year, so we wouldn't want to throw the whole year away. Let's err on the side of just discarding 2020.

```
In [11]: genres.loc[genres["Year"] < 2020, "covid_period"] = "pre-COVID"
genres.loc[genres["Year"] == 2020, "covid_period"] = "during COVID"
genres.loc[genres["Year"] > 2020, "covid_period"] = "post-lockdowns"

genres["covid_period"].value_counts(dropna=False)
```

```
Out[11]: pre-COVID      35
during COVID    16
post-lockdowns   16
Name: covid_period, dtype: int64
```



```
In [12]: genres.dtypes
```

```
Out[12]: Year                int64
Genre                object
Number of releases      int64
Gross box office (£ million)  object
Top performing title    object
covid_period           object
dtype: object
```

```
In [13]: genres["Gross box office (£ million)"].unique()
```

```
Out[13]: array(['361.3', '242.2', '154.1', '88.3', '78.5', '67.0', '60.3', '57.6',
               '57.1', '36.4', '30.8', '30.6', '17.9', '11.8', '9.3', '1.3',
               '0.1', '328.3', '312.1', '166.4', '83.5', '80.7', '75.3', '64.3',
               '60.6', '44.5', '42.7', '18.9', '12.3', '7.8', '2.8', '0.5',
               '<0.1', '48.5', '44.3', '34.8', '32.8', '31.4', '24.0', '12.5',
               '12.1', '3.8', '0.9', '0.4', '0.3', '0.2', '296.7', '65.1', '58.2',
               '40.8', '29.6', '20.6', '15.5', '15.2', '14.9', '13.4', '8.2',
               '4.5', '2.1', '1.5'], dtype=object)
```

There's a value "<0.1" which needs to be numeric. We could convert it to zero, but that would be misleading. It's also not the same as 0.1, which is a separate value in our data. Let's go for a middle value, 0.05.

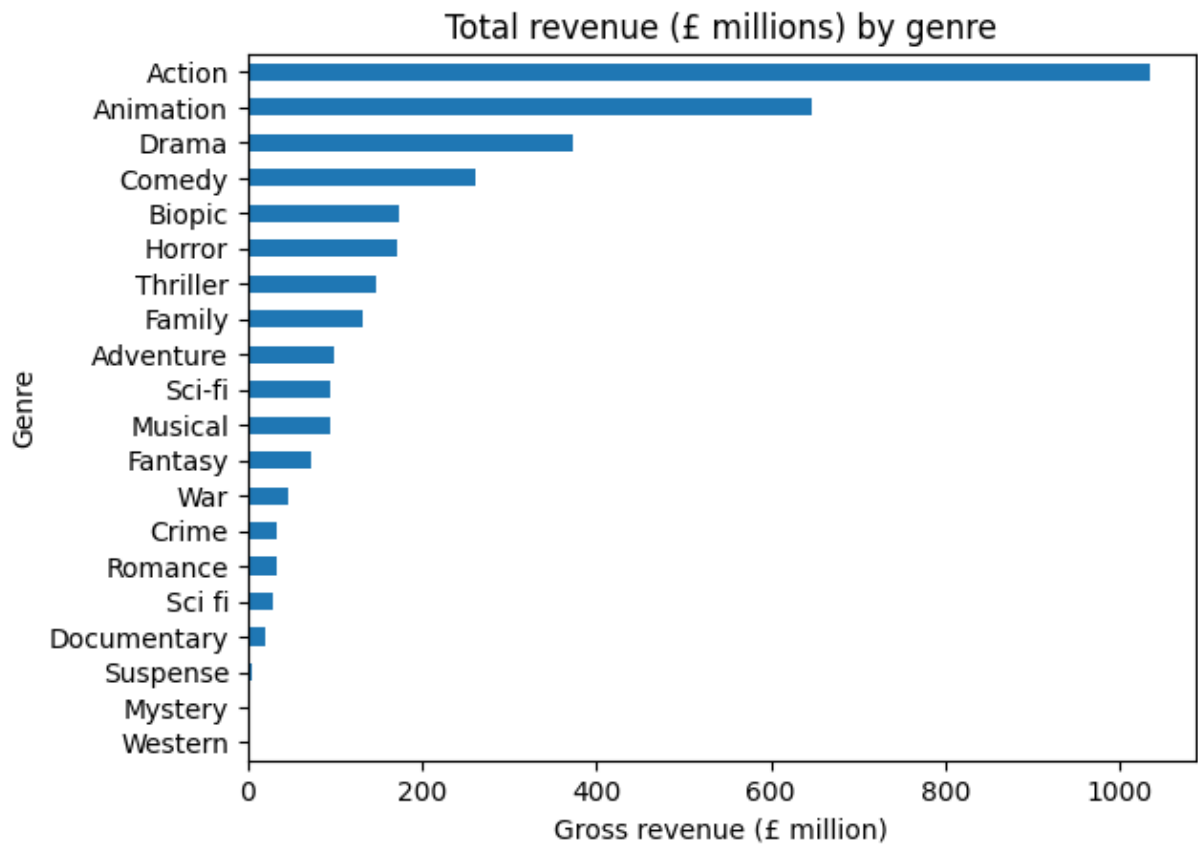
```
In [14]: genres.loc[genres["Gross box office (£ million)"] == "<0.1", "Gross box office (£ m
genres["Gross box office (£ million)"] = genres["Gross box office (£ million)"].ast
```

Let's look at how genres generally perform

```
In [15]: fig, axis = plt.subplots()

(
    genres
    .groupby("Genre")
    ["Gross box office (£ million)"]
    .sum()
    .sort_values()
    .plot
    .barh(ax=axis)
)

axis.set(
    title="Total revenue (£ millions) by genre",
    xlabel="Gross revenue (£ million)"
)
plt.show()
```



Generally people like films categorized as Action and Animation.

What about this distribution over time?

Let's look at each year's top genres.

By "top" let's look at revenue since "number of films" isn't a substitute for (perceived) quality.

```
In [16]: genres.groupby("Year").head(3)
```

Out[16]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title	covid_period
0	2018	Action	77	361.3	Avengers: Infinity War	pre-COVID
1	2018	Animation	46	242.2	Incredibles 2	pre-COVID
2	2018	Drama	246	154.1	A Star Is Born	pre-COVID
18	2019	Action	102	328.3	Avengers: Endgame	pre-COVID
19	2019	Animation	40	312.1	The Lion King	pre-COVID
20	2019	Drama	264	166.4	Downton Abbey	pre-COVID
35	2020	Action	36	48.5	Bad Boys For Life	during COVID
36	2020	War	3	44.3	1917	during COVID
37	2020	Animation	19	34.8	Sonic The Hedgehog	during COVID
51	2021	Action	52	296.7	No Time to Die	post-lockdowns
52	2021	Comedy	54	65.1	Free Guy	post-lockdowns
53	2021	Animation	38	58.2	The Addams Family 2	post-lockdowns

Looks like everyone loves an action film!. Interesting that Comedy wasn't in the top 3 before 2021. Perhaps people want more light-hearted relief post-covid?

Let's look at the entire distribution per year.

```
In [17]: years = genres["Year"].unique()

fig, axes = plt.subplots(1, len(years), figsize=(3*len(years),8), sharex=True)

for idx, year in enumerate(years):
    (
        genres[genres["Year"] == year]
        .groupby("Genre")
        ["Gross box office (£ million)"]
        .sum()
        .sort_values()
        .plot
        .barh(ax=axes[idx])
    )

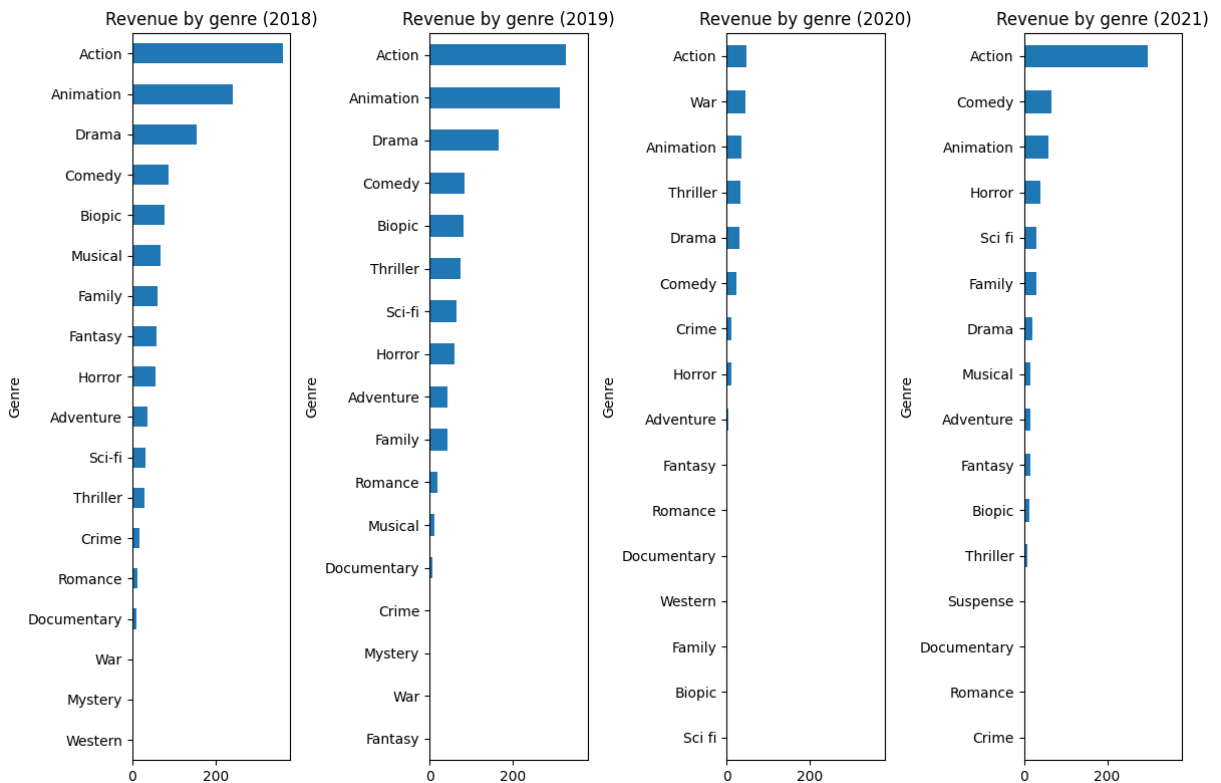
    axes[idx].set(
```

```

    title=f"Revenue by genre ({year})"
)

plt.tight_layout()
plt.show()

```



That gives quite a clear picture. Action films are the most popular, but there has been a rise in popularity of comedy films in 2021. Some theories:

- animations take years of effort and if animators weren't working at any point during COVID, that would have delayed the release of animated films in 2021
- comparatively comedies are probably cheaper to make (citation needed)
- people possibly prefer light-hearted relief in post-lockdown times
- the popularity of war films in 2020 is interesting

Let's test that first theory

```

In [18]: (
    genres[genres["Genre"] == "Animation"]
    .groupby("Year")
    ["Number of releases"]
    .sum()
)

```

```

Out[18]: Year
2018     46
2019     40
2020     19
2021     38
Name: Number of releases, dtype: int64

```

```
In [19]: genres[(genres["Genre"] == "War") & (genres["Year"] == 2020)]
```

```
Out[19]:
```

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title	covid_period
36	2020	War	3	44.3	1917	during COVID

Perhaps the film 1917 is responsible for that spike in popularity of War films, and less of an indication of changing behaviors.

Distributors

```
In [20]: distributors = pd.read_csv("distributors.csv")
print(distributors.shape)
distributors.head()
```

(44, 5)

```
Out[20]:
```

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
0	2018	Walt Disney	23.6	24	325.6
1	2018	Universal	19.5	40	268.5
2	2018	20th Century Fox	14.5	28	199.3
3	2018	Warner Bros	13.9	31	191.4
4	2018	Sony	10.7	33	146.8

Let's assign COVID periods again

```
In [21]: distributors.loc[distributors["Year"] < 2020, "covid_period"] = "pre-COVID"
distributors.loc[distributors["Year"] == 2020, "covid_period"] = "during COVID"
distributors.loc[distributors["Year"] > 2020, "covid_period"] = "post-lockdowns"

distributors["covid_period"].value_counts(dropna=False)
```

```
Out[21]: pre-COVID      22
during COVID    11
post-lockdowns  11
Name: covid_period, dtype: int64
```

```
In [22]: distributors.dtypes
```

```
Out[22]: Year          int64
Distributor         object
Market share       float64
Films on release    int64
Box office gross (£ million) float64
covid_period        object
dtype: object
```

Let's check that the market share column adds up to 100 each year

```
In [23]: distributors.groupby("Year")["Market share"].sum()
```

```
Out[23]: Year
2018     100.1
2019      99.9
2020     100.1
2021     100.2
Name: Market share, dtype: float64
```

Probably rounding errors, but let's check

```
In [24]: distributors[distributors["Year"] == 2021]
```

```
Out[24]:
```

	Year	Distributor	Market share	Films on release	Box office gross (£ million)	covid_period
33	2021	Universal	30.6	51	180.8	post-lockdowns
34	2021	Sony	22.2	52	130.8	post-lockdowns
35	2021	Walt Disney	21.2	63	125.1	post-lockdowns
36	2021	Warner Bros	14.4	80	84.7	post-lockdowns
37	2021	Paramount	3.7	16	21.8	post-lockdowns
38	2021	Lionsgate	1.8	23	10.5	post-lockdowns
39	2021	Entertainment One	1.4	12	8.1	post-lockdowns
40	2021	Park Circus	0.6	96	3.4	post-lockdowns
41	2021	STX Entertainment	0.6	3	3.4	post-lockdowns
42	2021	StudioCanal	0.5	36	2.7	post-lockdowns
43	2021	Other distributors (137)	3.2	549	19.0	post-lockdowns

Nothing odd, perhaps when other distributors were grouped into one category a rounding error occurred.

Top 3 distributors per year by market share

```
In [25]: distributors.groupby("Year").head(3)
```

Out[25]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)	covid_period
0	2018	Walt Disney	23.6	24	325.6	pre-COVID
1	2018	Universal	19.5	40	268.5	pre-COVID
2	2018	20th Century Fox	14.5	28	199.3	pre-COVID
11	2019	Walt Disney	37.9	24	506.8	pre-COVID
12	2019	Universal	13.9	39	185.6	pre-COVID
13	2019	Warner Bros	12.4	40	166.3	pre-COVID
22	2020	Entertainment One	15.9	30	49.4	during COVID
23	2020	Sony	15.6	38	48.4	during COVID
24	2020	Walt Disney	14.3	58	44.3	during COVID
33	2021	Universal	30.6	51	180.8	post-lockdowns
34	2021	Sony	22.2	52	130.8	post-lockdowns
35	2021	Walt Disney	21.2	63	125.1	post-lockdowns

This doesn't really tell a story, we should see it visualized.

One line plot per distributor showing market share over time would be interesting to see.

For that, our data needs to be one column per distributor and one row per year.

```
In [26]: (
    distributors
    .groupby(["Year", "Distributor"])
    ["Market share"]
    .sum()
    .unstack()
)
```

Out[26]:

Distributor	20th Century Fox	20th Century Fox*	Entertainment	Entertainment Film Distributors	Entertainment One	Lionsgate	di
Year							
2018	14.5	NaN	1.1	NaN	NaN	1.5	
2019	NaN	5.7	NaN	NaN	3.8	4.2	
2020	NaN	NaN	NaN	4.3	15.9	4.5	
2021	NaN	NaN	NaN	NaN	1.4	1.8	

Looks like some erroneous data crept in. Let's clean things up.

- 20th Century Fox appears with an asterisk
- Entertainment One is the same as eOne Films (source: <https://www.entertainmentone.com/about-eone/>)
- no evidence that Entertainment and Entertainment Film Distributors should be the same entity, we'll leave them separate
- finally, group "Others" together

```
In [27]: distributors["Distributor"] = (  
    distributors["Distributor"].replace(  
        "20th Century Fox*": "20th Century Fox",  
        "eOne Films": "Entertainment One"  
    )  
)  
  
distributors.loc[distributors["Distributor"].str.startswith("Other"), "Distributor"]  
  
distributors["Distributor"].value_counts()
```

```
Out[27]: Walt Disney          4  
Universal                    4  
Warner Bros                  4  
Sony                        4  
Paramount                   4  
Entertainment One            4  
StudioCanal                 4  
Lionsgate                   4  
Other                        4  
20th Century Fox             2  
STX Entertainment            2  
Entertainment                 1  
Entertainment Film Distributors 1  
Shear Entertainment*          1  
Park Circus                  1  
Name: Distributor, dtype: int64
```

Now let's return to our one line plot per distributor showing market share over time.

Let's limit it to those with data in all years since we're interested in change from pre- to post-COVID years.

```
In [28]: distributors_to_keep = (  
    distributors["Distributor"]  
    .value_counts()  
    .loc[lambda x: x == distributors["Year"].nunique()]  
    .index  
    )  
  
distributors_to_keep
```

```
Out[28]: Index(['Walt Disney', 'Universal', 'Warner Bros', 'Sony', 'Paramount',  
              'Entertainment One', 'StudioCanal', 'Lionsgate', 'Other'],  
              dtype='object')
```

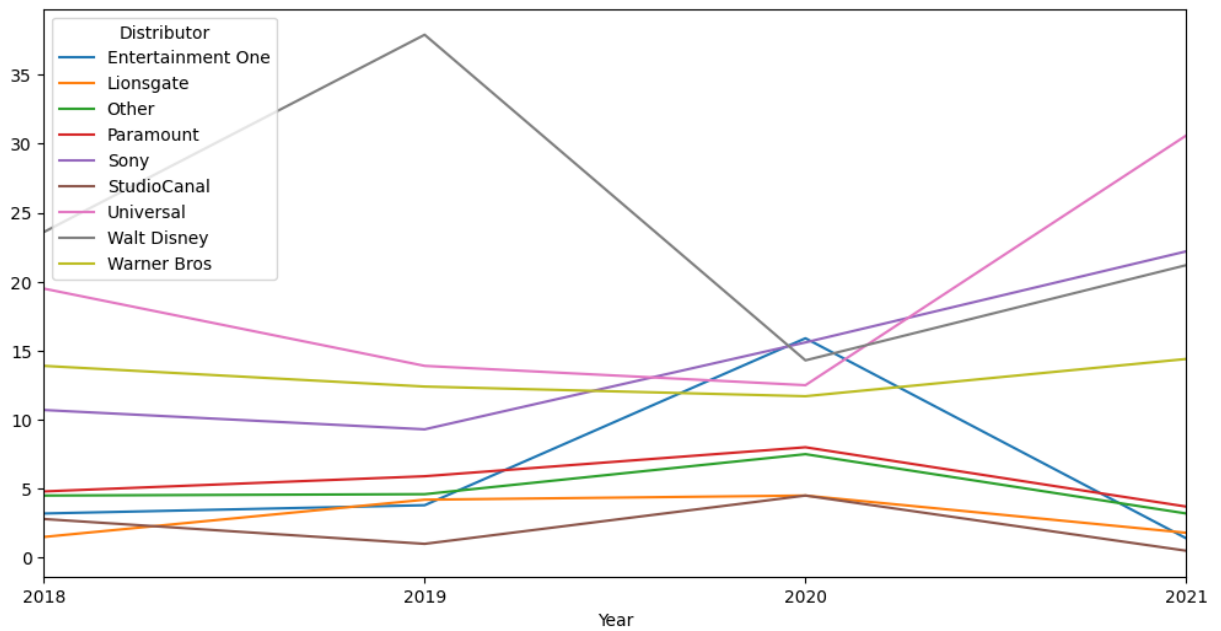
```
In [29]: distributors_pivot = (  
    distributors  
    .query("Distributor in @distributors_to_keep")  
    .assign(Year=distributors["Year"]  
           .apply(lambda x: datetime.datetime(x, 1, 1))  
           )  
    .groupby(["Year", "Distributor"])  
    ["Market share"]  
    .sum()  
    .unstack()  
    )  
  
distributors_pivot
```

```
Out[29]:
```

Distributor	Entertainment One	Lionsgate	Other	Paramount	Sony	StudioCanal	Universal	D
Year								
2018-01-01	3.2	1.5	4.5	4.8	10.7	2.8	19.5	
2019-01-01	3.8	4.2	4.6	5.9	9.3	1.0	13.9	
2020-01-01	15.9	4.5	7.5	8.0	15.6	4.5	12.5	
2021-01-01	1.4	1.8	3.2	3.7	22.2	0.5	30.6	

Calling `.plot` will give us one line per distributor over time:

```
In [30]: fig, axis = plt.subplots(figsize=(12, 6))  
  
distributors_pivot.plot(ax=axis)  
  
plt.show()
```



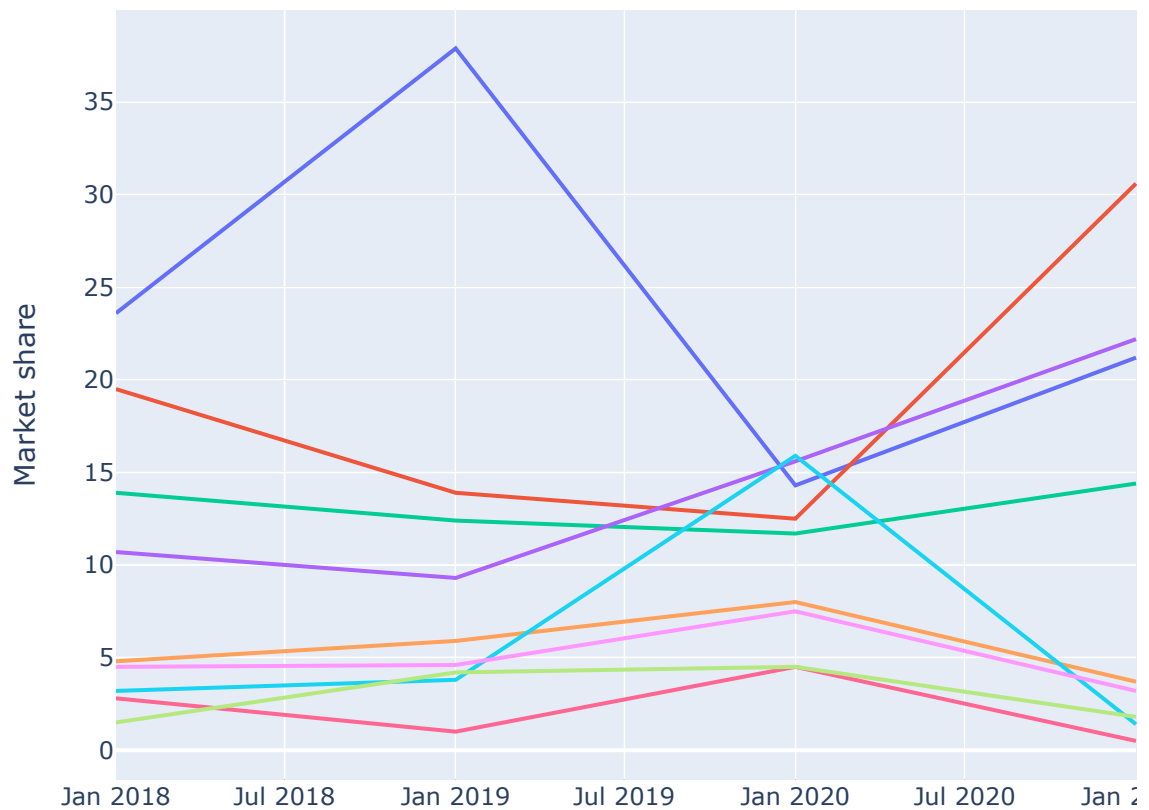
This is an instance where this would be better as an interactive plot so we could see which lines are which without having to look at a busy legend.

Let's look at the same plot with the plotly library.

```
In [31]: import plotly.express as px

fig = px.line(
    distributors
        .query("Distributor in @distributors_to_keep")
        .assign(Year=distributors["Year"].apply(lambda x: datetime.datetime(x, 1, 1)
    ),
    x="Year",
    y="Market share",
    line_group="Distributor",
    color="Distributor")

fig.update_layout(showlegend=False)
```



Let's also see the raw pivot table, so color plays no important role:

In [32]: `distributors_pivot.transpose()`

Out[32]:

	Year	2018-01-01	2019-01-01	2020-01-01	2021-01-01
Distributor					
Entertainment One		3.2	3.8	15.9	1.4
Lionsgate		1.5	4.2	4.5	1.8
Other		4.5	4.6	7.5	3.2
Paramount		4.8	5.9	8.0	3.7
Sony		10.7	9.3	15.6	22.2
StudioCanal		2.8	1.0	4.5	0.5
Universal		19.5	13.9	12.5	30.6
Walt Disney		23.6	37.9	14.3	21.2
Warner Bros		13.9	12.4	11.7	14.4

Some interesting stories here:

- Walt Disney bought 20th Century Fox, yet still dropped market share during/after COVID
- Most of the top distributors are returning to pre-COVID levels but the biggest names seem to have increased market share especially Sony and Universal

Possible theories:

- big distributors have the resources to withstand a pandemic and come back stronger, whereas smaller distributors are likely struggling to return to pre-pandemic levels

What does this mean for our white paper?

Just from these datasets there are already some angles formulating:

- admissions are making their way back to pre-COVID levels
- the rise in popularity of comedies post-lockdowns
- the largest distributors gain more market share post-lockdown